BE Semiconductor

Takeaways from our conversation with management

Overweight

BESI.AS, BESI NA
Price (26 Oct 26):€146.35
Price Target (Dec-27):€275.00

European Tech Hardware & Payments

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We hosted Richard Blickman, President and CEO of BE Semiconductor in an investor call this morning. After the announcement that Applied Materials would increase its stake in the company to 20%, he was pleased about the announcement given the increasing cooperation between the two companies in developing next generation advanced packaging technologies as well as positive about the order outlook, the co.'s competitive position and the potential for increased adoption of the company's HB and TC-Next technologies in the next 18-24 months in both the logic and memory markets.

- **Order momentum to continue:** Besi's CEO indicated that although AI has been the major driver of orders, there has also been a broader improvement across the semiconductor industry, resulting in order levels not seen for some time. Momentum was indicated as remaining strong. The co. indicated that Besi is seeing broad-based order intake and has now received cumulative demand for >215 hybrid bonding systems across 23 customers. The key question remains when hybrid bonding will move into volume HBM stacking. Current HBM orders are still largely for development, qualification and pre-production, while high volume manufacturing adoption in logic continues to progress strongly. The co. also confirmed that all three memory manufacturers continue to develop hybrid bonding for HBM, despite uncertainty around the precise timing of volume adoption.
- **On the memory hybrid bonding orders reported in Q3:** The co. indicated that HBM hybrid bonding systems are high-value tools, typically costing ~€2-3m each. We take away that the latest HBM order indicated in the Q3 report is a follow-on order from a customer who has already qualified the tool and is progressing toward pre-production and is ultimately being driven by requirements from a major end customer. The co. indicated that the other leading HBM manufacturers are also actively qualifying different HBM generations and preparing for future market requirements, suggesting additional orders should follow as these programs mature. The co. indicated that the fundamental advantage of hybrid bonding is that it enables better performance, lower power and greater interconnect density than conventional reflow processes. This is becoming increasingly important for data-center applications. Emerging architectures, including direct stacking of HBM/DRAM onto logic or accelerator devices, reinforce the longer-term direction of travel. Customer focus was indicated as increasingly on 2027-28, with significant developments expected over the next 18-24 months.
- **On logic hybrid bonding and adoption of HB at the key AI processor supplier:** The co. indicated that logic hybrid bonding also continues to progress, with improved yields and wider process windows. Besi is indicated to be closely aligned with the major logic programs and believes it will be able to supply the required systems when hybrid bonding moves to volume production. The co. indicated that customers are effectively also testing Besi's supply chain and its ability to reliably support a large-scale production ramp. Exact timing will ultimately depend on end-customer demand and technology qualification. As tool shipments increase and customer programs accelerate,

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Besi also needs to scale its service and support organization, as these increasingly complex systems require trained specialists to install, operate and support them. This expansion is therefore a gradual process rather than something that can be scaled up immediately. Besi has been involved in these programs for 5 to 6 years and has progressively expanded alongside customers, including with the AP7 program. The co. highlighted continued capacity expansion at leading-edge logic customers as indicated in the 3Q26 report, although the precise number of additional systems required will depend on end-market demand and customer production plans.

- **On competition:** The co. indicated that the emergence of additional competitors reinforces Besi's view that hybrid bonding will ultimately become a mainstream process. Besi believes that it remains the technology leader, with a highly reliable and cost-competitive process. Process experience is particularly important because customers need consistently high yields and predictable performance before moving HB into high-volume production. Besi has built this expertise over more than a decade through its own development activities and collaborations with Applied Materials and research/ application centres. Customers continuously benchmark competing solutions as pitches shrink, materials change and cost requirements evolve. Besi came out ahead in a major customer evaluation in Taiwan last year, which subsequently resulted in orders. Besi's CEO therefore did not seem particularly concerned about competition, although it expects customers to develop and qualify second sources for supply-chain security. Besi estimates competitors are around two years behind technologically, noting that it can take 5-7 years to develop a process that is genuinely production capable at scale.
- **On TCB products and competition:** Co. CEO indicated that TCB is becoming increasingly important alongside hybrid bonding. Besi's TC Next platform has gained traction, including its first repeat order for HBM stacking. The system offers tighter placement accuracy than conventional TCB and has demonstrated sub-10µm bond pitches at IMEC, versus current market standards of around 20µm. Importantly, customers do not necessarily have to choose exclusively between TCB and HB. Some applications will require HB, while others can continue using advanced thermal compression. TC-Next also supports fluxless processes, eliminating the need to subsequently remove flux. Besi indicated that it now has seven customers for the TC-Next platform and believes the 10-20µm pitch segment could develop into a sizable product line over the next few years alongside hybrid bonding.
- **On the additional Applied Materials stake:**Co. CEO indicated that when Applied Materials initially invested in Besi, only ~20 hybrid bonders had been shipped and HB was viewed as a superior technology but one with a relatively distant volume opportunity. The addressable market was therefore very small at the time, whereas it is now expanding rapidly. According to Besi, both companies provide complementary technologies. Besi supplies the bonding equipment, while Applied contributes materials, wafer/die preparation and cleaning expertise. As the HB opportunity becomes larger and more complex, the strategic rationale for closer collaboration increases. The establishment of the Singapore application centre and Besi's involvement in Applied Materials' EPIC Centre are examples of the relationship deepening incrementally. CEO reiterated what he said on the company's conference call, that the first EPIC Centre program would involve both hybrid bonding and TCB for a major US customer.
- **On China order strength:** Co. indicated that the China strength is broad based rather than being driven by a single application. Demand includes high-end smartphones, industrial/general semiconductor applications, 2.5D modules and some hybrid bonding development activity. Photonics is another important contributor, particularly because a significant portion of global pluggable optical-module production takes place in China. These end markets collectively explain the strong YoY increase in Chinese orders.
- **On gross margin progression:** The co. indicated that gross margins have remained resilient despite inflationary pressures and adverse USD movement, with Besi continuing

to operate around the mid-60% level. The CEO said that as hybrid bonding becomes more mainstream, there could be further margin opportunities, although profitability will remain a balance between pricing, product differentiation and ongoing cost reductions. The scale of the current ramp is significant, with Besi moving from roughly 1,000 machines (we believe this indicates all machines) last year to ~2,000 this year while maintaining its margin profile. The ability to sustain pricing reflects both the competitive positioning of its products and improvements in its cost structure. For 4Q26, Besi has guided to a 63-65% gross margin, consistent with 3Q, with the precise outcome primarily dependent on product mix.

- **On the strategic review:** The co. indicated that Besi last conducted a strategic review at the beginning of 2025, but the industry roadmap has changed materially over the subsequent ~18 months. This includes some delays in HBM hybrid bonding adoption, continued improvements in mass reflow and TCB, and a rapidly evolving advanced-packaging roadmap. Besi is therefore conducting another deep dive with key customers to reassess where the market is heading over the next 3-5 years. Key questions include whether the previous €1.5-1.9bn long-term revenue framework remains appropriate; how the revenue mix between hybrid bonding, TCB and mass reflow will develop; and what Besi needs to do to maintain its technology leadership over both the near and longer term. The review was indicated as being a 16 week process involving key customers and Applied Materials, with management expecting to reach conclusions around early January.

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Investment Thesis, Valuation and Risks

BE Semiconductor *(Overweight; Price Target: €275.00)*

Investment Thesis
BE Semiconductor (Besi) is a leading supplier of die placement, bonding and wafer level packaging equipment catering to the semiconductor industry. It is the market leader in hybrid bonding, a next-generation packaging technology that is poised for widespread adoption. AMD and Intel are known customers, with Nvidia and the major HBM suppliers evaluating the technology for next-generation chips. The indications remain that the technology is on the cusp of "take off".

Valuation
We value Besi based on '28 estimates as Besi's stock is mostly driven by its longer-term dynamics – namely the adoption of hybrid bonding technology in high-volume applications. In view of the delays we reduce our multiple to the midpoint of its trading range and value Besi on a 33.5x multiple (mid-point of the inter-quartile range) on '28 EPS forecast which implies a €275 target price. Although this is a considerable reduction from the prior €342 price target, it still provides a considerable 70%+ upside to the current share price. The stock will require more supportive news flow to get to the target price because though the orders/sales will likely remain strong in the near term, the market focus is on Hybrid bonding adoption.

Risks to Rating and Price Target
Key risks to our rating include:

- **Economic risks:** The ongoing macro risks associated with US tariffs are not factored into our estimates, rating or target price as impacts are uncertain. However, negative impact on demand would negatively impact the stock.
- **Competitive risks:** Besi's valuation premium to its peer group is a function of its market-leading position in next-generation hybrid bonding. If competitors can catch up with their technology, then risk is that the valuation premium reduces because long-term growth will be slower.
- **Risk of delay in hybrid-bonding adoption:** Though Besi has sold a large number of tools in hybrid bonding already, timing of adoption of any new technology takes time and is dependent on various factors such as cost effectiveness, timing of the launch of the new products using the technology and on the overall market environment.

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Date	Rating	Price (€)	Price Target (€)
09-May-25	OW	106.10	121
28-Jul-25	OW	117.10	135
21-Oct-25	OW	145.20	147
27-Oct-25	OW	147.05	160
13-Jan-26	OW	162.15	172
23-Feb-26	OW	184.50	204
21-Apr-26	OW	227.00	230
27-Apr-26	OW	252.20	266
15-Jun-26	OW	317.00	342
19-Oct-26	OW	155.55	275

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage May 08, 2025. All share prices are as of market close on the previous business day.

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